

Subtitles of PoS videos

https://www.youtube.com/watch?v=ulqv_tX_uAY&t=1s

hi

i'm here today to talk about the short
essay written by duncan watts
about why everything is not obvious
so it's it's kind of an overview of of
the paper and i'm going to talk about
the key questions that are asked
and the kind of answers that that duncan
watts provides here
um and and the

Paper Summary

four primary questions you could say
that he's asking in this essay here is
why do we often think that social
science is obvious
after the fact? what is it about social
science it seems different
from quite a few of the other sciences
for instance physics or engineering
um you could talk to an every
layman on on on the street and and they
will have an opinion and they will think
that they know something about
unemployment or management or strategy
and then often results that we get might
seem obvious
and then sort of the point one of the
key points of this article is then to
argue and show
well it actually it's actually not
social sciences are not
as obvious as it might seem um they are
particular and are different in certain
ways and for instance
the natural sciences and just also short
indication of what's then the solution
what
how do we solve this particular
challenge here
so it's it's a really foundational
philosophy of science perspective that
also has implications for how we can

actually do science
and in particular question three but
actually also so the overall essay
very strongly relates to the Nelson
paper that that is also on the
curriculum for
for this class here okay so to
try to create a contrast sort of quite
drastic and radical contrast to what
social science is about let's
try to think about an example from
physics that is
at least to me quite mind-blowing so we
have a guy called Cavendish
he worked in his own lab in England in
the late 18th century
and he built a device sort of this spec
here
more or less a wooden device but it had
some some
metal balls and some strings and things
were attached to each other
and basically what he was doing in his
own basement
in England in the 18th century was
measuring the weight
of the Earth he was about one percent
off it later turns out with
modern science I think that's reasonably
okay
so again sitting in his basement being
able to measure the weight of the Earth
and since we knew Newton's laws of
gravity
he could also infer the weight of all
the other planets
and that's fair to say that these
results were not obvious at all if you
talk to
the regular person on the street they
would not
have been able to figure this out they
probably wouldn't even have been able to
figure out that
you could measure this weight and you
could infer these things
and then you could ask yourself what is
it that we social scientists or in

business administration what what can we
do with
in our lab or in our uh sitting in our
office trying to figure things out
it's certainly a very different field
that we're in
um and the sort of one point that is
often made that
should be obvious that social sciences
should actually be less hard than
physics or like literally that's the
expression that we use rocket science we
can
send rockets to the moon and and surely
that's more difficult than figuring out
human behavior the moon is pretty far
away
and that's sort of the puzzle we and
what is it about social science it seems
particular
let me try to get to this from from
another angle
um trying to illustrate this obviousness
so following the second world war
there was a really large-scale effort to
try to investigate
what it was like to be a soldier so they
were interviewing six hundred thousand
soldiers
um it's quite a large case study and one
of the key questions was who handles
life as a soldier better
and surely people from the countryside
they're used to physical exercise
they're used to being out in the open
maybe sleeping out in the open handling
different
materials and just living a quite a
rough life
and then whenever the main researcher
here he presented this story this was
also how he
told it and and people in the audience
would often go
surely you didn't have to interview 600
000 people to get to that result
but actually and that was a trick that
he often pulled with the audience it was

actually
people from the city it was it was
soldiers that had grown up in the city
that were
the best that could cope with being a
soldier the best
they're used to being in a uniform they
were used to working in a hierarchy
they were used to working at a specific
specific time schedule doing what others
tell them to do
um so this is a typical situation
in social science where both kinds of
answers completely opposite answers
could sort of seem obvious uh when we
are told about them
um here is just a slightly longer
explanation and a link to the story
about the
the american soldier and unobviousness
and we can think of other
examples here that from psychology that
it might
seem obvious that if you want someone to
say yes to a small request
giving a bad reason is surely better
worse than no reason at all
or if you want to be happier count as
many blessings as you have
count how many things that makes you
happy as possible
surely that's better than only counting
a few
we can think of other examples as well
right
surely but again um as as research has
shown
it's actually the case that giving a bad
reason
for in this case the research was done
on standing in a queue of getting to a
copying machine
and and and if you then gave a bogus
reason for why you caught in line
in front of the queue you said i need to
make copies well
surely everyone here in the queue needs
to do that it was a cue to the copying

machine but actually
it's just as good to say this bogus bad
reason as it was to give
a good reason because i'm in a rush and
a slide here
illustrates some of the other examples
and one could come up with a range of
other examples that that
what might seem obvious is not
necessarily so
um but again this is a particular
situation that we're in that that
we if it it seems obvious and we should
be
but we are much better at rocket science
a field that does not seem obvious
we can send rockets to the moon or even
people to the moon
while figuring out what movie is going
to be a hit
or whether we should change the
unemployment rate with how much or
whether an advertisement actually works
those are really really difficult
questions and and these seems to be
basically much harder than rocket
science at least we're not making as
much progress with them

Why It Appears Obvious
so question is and that's also what
what's then is writing about here why is
it that social science seems obvious
what is it about
this field that is quite particular and
and for one
is that it's it's it's familiar so we
whenever we work for instance in
business administration whatever
bachelor project you'll write or master
thesis you might write about how
motivated people are or stock market or
company strategy and these are all
topics that we
that we live we read about them in the
newspaper
we we think about them and we might have
opinions about what is good management

how should a manager behave
surely this is the right way to do it we
don't in the same way live in a world of
atoms or black holes or quants or
whatever they are dealing with in
physics
so that that sort of one element that
can lead to
our signs being particular it's it's
familiar we face it every day
so surely we know something about it
but it's also related to human bias i'll
be talking more about in them
in a later lesson that we tend to after
rationalize after the fact
we tend to when we've been presented
with some information or solution to
something we tend to think
surely that's the way it was supposed to
be i could have told you that in advance
even though that we might not have been
able to do that
so this after rationalization also makes
things appear more obvious
after the fact and then i think also key
element that common sense
is usually quite effective in
everyday life
we get i mean it's quite efficient that
we know a lot about how the world works
or when we enter a room when we deal
with objects
we know a lot about we're quite
effective in in the way that we can cope
with everyday life
um on the other hand common sense also
has an answer to almost everything
birds of a flock flock together but
opposites attract
it's bad to sweat the small stuff but
good to be detail-oriented and we can
sort of come up with
all sorts of common sense answers to
lots of life's
predicaments and challenges so
so what's here lists a number of
elements that
shows how social sciences and business

administration in particular
has a particular nature and can seem
obvious because it's familiar because it
is about stuff that we deal with on an
everyday basis and and makes us think we
know what the social world is all about
but and that's really also then one of

Why It Isn't Obvious

the key points

it's not it's not obvious predictions
and explanations are really really hard
in the social sciences because we live
in such a complex world
when we try to explain whether a company
strategy is going to be successful we
have to take into account how the
employees are going to
develop it and how they're going to
execute it and how competitors are going
to react to it and whether there's a
financial crisis going on at the moment
and how customers are going to react
and and etc etc they're all these
different levels and they all
influence and interact um
which makes the social world extremely
complex in a little sense of that world
and we'll get back to that in a in a
later lesson as well
but humans are also particular in the
sense that they can change the
behavior and change their minds if we
figure out how the planets
they evolve or go around the sun
in the solar system i've got to change
behavior tomorrow and they're going to
continue
in the same direction no matter what we
figure out and what explanations we
we create while human beings if you tell
them about
a certain advertisement strategy and why
companies do what they do
customers might react differently to
while having that awareness
um so so that also makes the social
science world quite particular and

particularly hard
uh because human beings are not static
and just continue in
the absolute same direction as many
elements in a in a physical
world would do and then also
we often don't know the counterfactuals
we often don't know what
what could have happened but didn't um
Taleb has a good example in his book the
black swan about
the greatest the person that could have
had
been the greatest hero and could have
had the greatest impact
would have been the one that had come up
with the law saying that all pilot doors
have to be bulletproof and locked which
could have enabled
avoiding the 911 terrorism incident in
the United States
um if there had been such a law but
there wasn't but what of all the
different
laws and rules and activities that have
been put in place that
that where we don't know what it
actually prevented
again we live in a very particular kind
of social world that it's complex human
beings can change the behavior
we don't know exactly what impact our
activities and actions have
um because we don't know what what could
have happened

Summary and Future Lessons

here a few more quotes from from the
paper on why social science is hard and
different from physics
so that was the the essence of the paper
that i want to present i wanted to focus
on well what are the key questions that
are being asked in this
quite short essay um
and i want to give this overview and
this understanding of social science i
think is quite

important for understanding the channels
that we face and
and the situation we are in in our
discipline
it has some consequences for how to do
science which i'll get back to in later
lessons
and then in particular i think it's also
a good thing to read this
paper because it relates so directly to
another paper Nelson
that we're going to get back to in in a
later
lesson that so to further dive into what
is it about
social science why is it it's a
particular field
and and what philosophy of science
implications does it have for the
specific choices that we should make
when we're doing a bachelor project when
we're doing a research project or
whatever challenge we're we're facing

<https://youtu.be/EdCyxonpTxQ>

now it's time for an introduction to
what an experiment
actually is so
what is a randomized control trial
that's the same thing as an experiment
it's just
a randomized control trial it's sort of
the more technical term that makes
explicit the three different elements i
want to
get you as give you a sense of why i
experiment the gold standard in terms of
establishing causation
and later on in the live session we'll
get to internal or external validity
okay and and the aim or the learning
objective is that you will be able to
assess the quality of an experiment
whether it's in business administration
economics more in general or whether
it's in all sorts of other
fields and medicine or whatever whatever

it could be you'll
learn sort of the essence of what to pay
attention to in order
to be sure that this is an actual
experiment

Importance of Experiments

so this is the question that we also
grapple with in the form of video
how can we be certain that in fact is
due to one specific variable
how can we be certain that it wasn't
just a random um correlation or that
there aren't other factors that we
haven't thought about
or the variables that we hadn't thought
about that actually are the key
that are key and as the chief economist
at
Google said there in 2016 to find out
what happens when you change something
it's necessary to actually change it
that's really the essence of the
experimental approach
is that in order to be sure that
something has an effect
you need to try this effect out and see
what the results are and and i also
refer to someone from google just to
highlight that
experiments of course isn't just
something that we as researchers carry
out
um these numbers are even now a couple
of years old so they're probably
outdated and much higher now but at this
point in time
every year um google and amazon and
netflix etc ran
thousands and thousands of different
kinds of experiments so in that sense
you've also been exposed to a lot of
experiments just by being online
because netflix constantly tries out
different ways of presenting their
information to the consumer in order to
see
how do we get people to consume the most

um Jeff Bezos from amazon even says our success at amazon is a function of how many experiments we do per year per month per week per day so they do this all the time and again of course they also do it in a in a proper way there's no reason to sort of do and a half-baked experiment that doesn't work you have to do it in in the right way and another illustration of this comes from from this sort of small article here an experiment experience that ebay had a couple of years ago um basically they had the experience that the more they changed and ad expo the more they used ads on google for instance the bigger the change in the more they also sold their products so basically an ad in google is you you probably know this or you know all the other search engines if you type in a word there will often be an ad associated to it so the top link here is actually not sort of a and a link provided by your google search as such it's a sort of an additional ad feature that is on top of it that google is making money on and what ebay was experiencing is that friends every month or every quarter they were sort of increasing the number of ads they posted and the more um ads that were exposed to human searches well the more products they solved so fantastic their ads work right well it turns out they actually got sort of an unplanned way of testing this because in relation to a negotiation they had with another company they wanted to show well how big was the effect of this

this ad and so they shut down the ad feature and what effect was there well okay this link says absolutely nothing that might be overdoing it a bit but as you can see visitors via the paid link versus visitors via the unpaid link the moment the ad feature stopped the number of visitors that went to the unpaid link that is clicked on this one which doesn't give google money it's only this one here that does just increased a lot so um so so basically what ebay and this is was at this point in time also a very large company run by very very qualified people but they for years did not realize that their ads did not give them the value that they thought it they would simply because they hadn't run an experiment to test it so just a sort of very sort of business oriented example of the need to run an experiment even in situations where you think you have a pretty good idea of what's going on because you could after all be wrong it might appear obvious as Duncan wants to say but it wasn't after all so what do what

How to Conduct a RCT

do we do what do companies like amazon and google et cetera do when they actually run an experiment so the essence is as we often use the the medicine sort of a headache build example because it's it's quite intuitive you want to find out if a headache pill is effective so you find a number of people let's say 100 people that have a headache all have to have sort of the reasonable same kind of headache you then randomly divide them into groups one group gets the pill one group doesn't get a pill

and in that way we can then see over time for instance six hours how many of the people in this group that got a pill have a headache versus how many in the group that didn't get a pill it turns out that the reason why i have three distinctions here is that it turned out that there is something called a placebo effect just getting a pill makes you think that there might be a positive effect so just the interaction with the doctor or the nurse or even the pharmacist can have a positive effect so it turns out the placebo effect is actually quite substantial in terms of reducing headache

um it does um in fact have a substantial effect what we're interested in is the actual chemical effect of the content in the headache pill that's what we're really after we don't want to sort of identify a placebo effect

so this is the way we do it we have a number of people we randomly allocate these people into one group another it should not be i have the most headache therefore i need to get into group x or my headache isn't that bad or i don't like pills so let me put it down here then we could certainly see other kinds of effects

that if we end up having the people with the most headache getting pills well that might also be most likely to get their headache reduced by a pill so we want these groups these three groups here we want them to be as absolutely similar as possible and as indicated this little picture here of course people shouldn't know that they're getting a placebo pill or that they don't get a pill or i mean this is ideally people don't know which

condition they're in because again if they know that they're in a certain kind of condition they might adapt their behavior accordingly and only one variable should be changed so ideally these people in these three different groups they behave completely identical just because you get a headache pill you don't go out and play soccer and get the football in your head and thus increasing your risk of getting a headache i mean you should sort of have the same kind of behavior in all groups before and during and after the experiment in order to get as clean a test as possible we'll see a bunch more example of this as we go along this is just to really identify the the essence of it it's a randomized trial of random allocation it's a controlled trial so we have to control that all these groups are sort of similar in as many ways as possible and it's a trial or it's a randomized control treatment there's only one thing that is being done differently so that's the wether um RCT it's not an RCT that's what the RCT is is coming from RCT and let's take another example that's from the Ariely text in the curriculum so Ariely wanted to find out if seeing the fruits of our labor makes us more productive so he did an experiment where he took a bunch of students at harvard and randomly allocated them into two groups in one group well in both groups they had to build lego bionicle figures but in one group the um the

figure was disassembled that was taken apart
right after they had finished building it so in this specific
uh group right after you've you've finished building your your bionicle
someone is standing in front of you and taking it apart it's very clear that
there's really no point to building it in the other group what it's called a
meaningful group it actually just means that no one took it apart
so as similar setup as possible they both build lego bionicles they both
get the same amount of money then the question is how many are they
willing to build as you can see you get three dollars for the first
270 for the next 240 to one etcetera so for how little money are you willing to
build the next bionicle and it turns out as you can see
in the meaningful group they were willing to build 11 in the
in the other only 7 so here we have another randomized control trial where
we have a group of people randomly allocated to one group or another again they're not
being asked so would you feel it annoying if someone disassembled your
bionicle right in front of you and if they say yes they get into the
meaningful group no we don't want this kind of selection we
want a completely random allocation of people into one or the other group
the condition should be entirely similar this shouldn't be sort of be
happening in in sunshine while this happens in rainfall
all other factors should be the same only one variable should be changed
the fact whether it's disassembled or not
and again of course we have to remember

this is an average causal effect on
average
they built 11 and 7 here of course there
was a distribution of
of number of bionicles they built across
individuals
so just sort of to really reiterate all
these elements here randomized control
trial
we have an allocation to each treatment
so we have a group of people
that we then randomly allocate to each
group notice that the people in each
group they're not necessarily identical
i mean one person can only get into one
group you can't
split and copy one person and then make
sure that
they're completely 100 identical we
can't clone people
organizations or whatever we're starting
like that but we can assume that if we
had a group of people
and we then randomly allocated for
instance via coinflip into one of the
other group
they are as similar as possible they're
not going to be identical
but they are as similar as possible one
group plays
it has a one kind of bionicles condition
the other one the other one
and we can then see the results but the
idea is that if you
because you have this random allocation
to each group then these groups are as
similar as possible um
and for instance you can think of it in
terms of gender let's say you have 100
people and you divide it up into
by by a coin flip into one of the other
group if you start with 15 men and 50
women
most likely you have about 23 24 25 26
or something of each in each group
it's not going to be completely
identical number
in terms of gender but it's gonna be

very very similar just because of
of how randomness works when it
accumulates
here i have a picture of two classrooms
because we get to a classroom example a
little bit later
but just to show that if you run an
experiment in a room or
wherever you run your experiment it
should be completely identical
um the setting should be as identical as
possible
and then which then enables there to be
only one treatment there is only one
thing that is
different from one setting to the other

What Is NOT a RCT

so let's try to
think about this a bit um
and and think of this example here it's
very famous example it was also
documented and showcased on danish tv
it's the marshmallow example
so it's an old study from from back in i
think the 70s
where four-year-olds were offered one
marshmallow now or two marshmallows 15
minutes later
so as you can see there's a kid sitting
in the room there's a marshmallow in
front of them and they can eat it now or
they can wait 15 minutes and they would
get two
this is a randomized control trial you
can if you want to just pause the video
and reflect on it
and consider whether it's sort of it
fulfills the criteria we have oops there
was one boy who couldn't wait
um who knows maybe they
weren't telling the truth when they were
promising two marshmallows later so
better eat it now right
um in this situation
yeah better hide the plate then no one
knows you ate it um in this situation
it's not a randomized controlled trial

because there is no sort of
everyone is getting the same treatment
everyone could get two
marshmallows later so they're sort of
self-selecting into
either eating right away the one or
getting to getting an additional one
later on
there was not a random allocation in
terms of um
of who gets how many marshmallows so
it's an exact identical setup and people
sort of self-select into
whatever outcome they end up with you
could have done an experiment by saying
so what is the impact of having to wait
seven and a half minutes versus 15
minutes that would have been
saying a direct comparison let's take
the exact same setup
put kids into a room some kids wait
seven and a half minutes other kids wait
15 minutes
let's do a comparison of how many
marshmallows are being eaten right away
that would that would have been a way to
say it's an experiment so we need to
make sure there is a randomization
element
that's what this example is is
highlighting
um sort of the randomization element we
need to randomize people into different
settings so there can be a variance of
treatment so again also the
treatment is not identical there is not
sort of a thing that is different
across the situations okay so let's
take another example and read about this
basically a
reference to the bible and what could be
the first documented randomized control
trial if it actually is one
so again you can pause the video and
read the snippet that i've inserted
there and think about it
um i'll go through it here so the
essence that

the way that is being described in the
by in the bible is
Daniel is a fellow who doesn't want to
eat meat and insist that it's healthier
not to eat meat
king was interested in this okay how
about
we test your servants for 10 days um
or Daniel's is proposing this so we
should test your servants test the
servants for 10 days
give us nothing but vegetables to eat
and water to drink and then we compare
our parents with that of other young men
who eat the royal food
and then to see what's going on so the
king ends up comparing the four non-meat
eaters including Daniel
with other meat eaters and then after 10
days
after this 10-day treatment they are
then compared and then
to assess are those at a meat do they
look healthier than those that did not
eat meat
and and and here's there seems to be a
treatment
meat or not but
concerning the allocation of people into
each group
it then there is self-selection in the
sense that daniel who came up with the
proposal
and who doesn't eat meat he is himself
in the non-meat eat
group so so maybe daniel is not eating
meat because he
already is healthy and can sort of um
get by by not eating meat so maybe sort
of the effect is coming from somewhere
else in that sense that
the two groups are contaminated so to
speak because there is not a complete
random application from one
in in one group and the other group and
it could be the case
that um the people that ended up in the
non-meat eating groups they were already

healthier in advance
they were already maybe the kind of
people who sort of could
eat enough vegetables and still look
healthy so
we need this random allocation and
that's why this is not an actual
experiment even though it's pretty close
and pretty impressive that
according to the bible this happened at
one point
okay let's do one more example um
researchers interested in the impact of
rewards on behavior
so in one group 50 people are said to do
a task as many times as possible and
receive one kroner per task
the other group 50 people are to do the
same task as many times as possible and
are told they will receive
a 1.50 kroner per task and the goal is
to solve 50 of them
pause the video if you want to reflect
on it maybe you already are at the
answer
two things are going on here we are
comparing one kroner per task with one
and a half kroner per task
but on top of that one of the groups
we're told that there is a goal
involved you should solve 50 tasks goal
setting behavior
does have an impact as you know from
organizational behavior so
the groups are not completely identical
this suggestion should also have been
given to the original group
in order to compare the two settings
and of course those in one group
shouldn't all
shouldn't know what is happening in the
other group because you might be annoyed
about
landing in the one known of a task if
you could have landed at one and a half
and that was just bad luck
and this annoyance could then lead you
to performing more poorly or what sort

of whatever i think that could be there
so
as highlighted here we should not we
should never really know what is a
treatment in other groups so if you
participate in the
in the COBE lab in the in our social
science lab here at our university at
bss
that's why you should not sort of know
what treatment you're getting or what
treatments others are getting because
that could lead to
some kind of hawthorne effect where you
know you're being observed on this
particular variable and
and maybe you'll adapt your behavior
according to it it's a challenge of lab
experiments that we'll get back to

Key Elements of RCT

so key elements that should be a
randomization process
whatever group of people you have you
should allocate them randomly into one
or the other group
next time we'll get back through in the
next video we'll get to thinking about
in the Ariely experiment with
bionicles it was actually only male
students from howard
that participated so is that then still
completing the
criteria of randomization second point
is you should manipulate the treatment
should have been
only one variable comparison with
control and treatment there should only
be one thing that is different
and then it should be a control setting
so everything else should be the same so
you should really make sure that if you
if you have two different groups at two
different times that you try to make
them as similar as possible
of course nothing can be completely
similar if you run one part of the
experiment at 10 o'clock and the other

one at 11 o'clock
well people at 11 will have gotten out
of bed a bit later and maybe the sun is
shining a little bit more although the
room is lighter
i mean it should basically and in
essence be the same
but of course even the air that is being
breathed in the room is not completely
identical
you can never expect sort of different
groups to be absolutely completely
identical
what you're aiming for is that it should
be as identical as
absolutely possible this was the essence
of a randomized controlled trial we'll
get a bunch more
examples and illustrations and also of
different kinds of experiments
in the coming videos and sessions but
these were
the basics as they are

https://youtu.be/Wz-1b_B6lnU

n this video i will talk about
correlation versus causation
these are concepts that i'm sure you've
heard before i'll try to
dive a bit deeper into them here in this
in this video so idea is to get a sense
of what do these concepts mean and how
and how are they different um but also
talking about what does it actually mean
for something to cause something else
and
should we even think of single or
multiple causes and what is a cause
in the end the meaning of the phrase
is that we often claim that something
causes something else so the
meaning can be sort of instantiated in
many different ways we can say something
leads to or brings about or
harms impact reduces something increases
something else or decreases something
so we haven't we use it often in

everyday language and say that something causes something else and this is what we'll want to get a sense of and this will then lead into part two of this lecture where we will dive into the golden standard for establishing causality which is experiments or randomized control trials the same thing

Correlation Fallacies

okay so we have this idea and i think we have this notion that we know that correlation does not equal causation just because some things are correlated does not mean that they are sort of causally linked to each other so this title mission page has a bunch of examples of weird stuff that is just extremely highly correlated so the consumption of margarine is 99% correlated with divorce rates in maine and it's not like people get divorced because they consume too much margarine or the other way around um there are a bunch of other examples it's just because there's so much data in the world that in the end a bunch of stuff will be linked what we want to get a sense of is when can we be certain that there is a causal relation because what is going what is happening in the human mind and we'll get back to this in lesson eight when we talk about Kahneman's text is that we as human beings have a tendency to to see stuff being linked in a causal way that's how we make sense of the world we link things and that's why we sometimes end up linking things that shouldn't be linked because we have this natural tendency of of linking two different things together in this case with margarine and divorce

rates it's probably so obvious that we should be too worried about it but there are others more sophisticated and more subtle situations than we'll try to address where this is not so straightforward but again we are after the cause a causal effect so sun can cause more consumption of ice cream it can also cause people to get sunburned but of course sunburn doesn't cause more ice cream to be eaten or ice cream an increase in the consumption of ice cream doesn't cause more sunburns but these things are correlated the more ice cream that is being eaten in denmark it's correlated with how um how much of a sunburn that individuals are getting but again this is because of another causal factor and that's really a key thing that we'll try to get a sense of here is that we need to be sure in the way that we design our research that we can be certain that it's the actual causal factor that we've gotten a hold of and not some more sort of spurious correlation whether something is correlated but it's because of another factor and you can maybe put the video on pause and just read this reread this piece of text that's from the Barrowman from the curriculum i want to sort of highlight the essence of it because it illustrates how we can sometimes identify something that clearly seems to be related to the but but actually might be sort of uh hiding another underlying causal factor okay so this is a very famous example of a psychologist Gottman who interviewed married couples and based on these interviews based on these conversations he had with these married couples he apparently had a 94% success rate in

terms of predicting whether these couples would get divorced within three years
so calling it a success rate and a prediction might be sort of a little bit off
so but he was very good based on these conversations because he could identify how sort of how much content and criticism and how defensive they were when they talk to each other these married couples
he could then use that as a sort of information
to get to his prediction but this is then often as Barrowman points out often misinterpreted because it's not like you could then divorce proof your marriage and change the way you communicate and then make sure you don't get divorced because the issue is that these factors of communication are not necessarily the causal factors that create the divorce
it could often be the case that it was actually another underlying factor whatever that could be of disagreement or or unhappiness that caused the poor communication and also caused the divorce it's basically the same thing as we saw with the ice cream example
that these two are strongly correlated but in the end there is another causal factor that is causing both of them so the point here is that um we can't be sure that the outcome is due to these communicative behaviors and we certainly can't be sure that changing these communicative behaviors would actually change the outcome and it turns out that Gottman actually wasn't that successful in sort of making sure that people didn't get divorced um so again what we are after is that we

want to
create a research design where we sure
that we've identified
actual causal effects and not what we
can call a confounder
confounder is with something sort of
interacting or
mixing or mixing something up and in
this case it could be
that the communicative patterns
are not the actual underlying core cause
but they sort of
confound they confuse our interpretation
of what is actually going on
so this is something that we want to
learn using
uh our insights into research design
that will be generating
to become better at making sure that
there is this causal effect that we have
another challenge that we already heard
about in lecture one was is this thing
about selection we have to be sure that
what we're seeing as
a sort of a causal effect it's not just
because we see
a biased um selection of data it's not
because we only see
part of the data we only look at
successful entrepreneurs or only look at
the planes that actually get home
um in the second world war that then
leads us to a certain kind of conclusion
for instance of how to become a
successful entrepreneur
so that can often then also be an issue
that um it's because we only look at a
subset of the data that we can end up
thinking something is causally related
even though that it's not we might end
up concluding that all successful
entrepreneurs
sleep at least five hours per night but
but the unsuccessful entrepreneurs
probably also sleep five hours per night
so it's not really an actual causal
relation
in a meaningful sense and the reason why

The Role of Media

all these things about causality is
is really important is highlighted in
this slide
which i'll try to really go through in
detail because it's quite important
the reason is that scientists and
especially the media has a tendency to
oversell a certain story okay so what
are we looking at here
on sort of this you could call it the x
axis from left to right
we have a categorization of the strings
of causal interference
how certain can we be that what we've
identified is not just a correlation
but it's actually a causal relation as
you can see this is not sort of
either it's true or not it's sort of
continuum
to the left there are lots of problems
with the research design and we can't be
certain that this is actually causality
the more we to the right we can say okay
this is very very likely that we've
identified a causal effect
what we see when we look at the academic
studies and this is then
the group of researchers who've looked
at the 50 studies
published in this journal plus one that
have been shared the most on social
media
so they found the 50 studies that have
been shared in total 1.3 million times
so these have really been shared a lot
these have been read by many people
um i've looked at these 50 studies and
then a group of
of of experts have rated these studies
and assessed
well what's the strengths of the causal
interferences
that these studies could make as you can
see only four percent
only two of them were really in this
very high category

one was in the higher category but a big bulk of them were in sort of the low or the moderate categories um meaning there were certain kind of problems in the research design that meant that we can't be 100% sure that identified an actual causal effect if we look at the studies and themselves well how what what cause language do they use in these 50 plus one article so what language does the authors themselves use this the first row is outside experts that have re-categorized what this research design and the studies actually could do the next layer here is the researchers themselves who've authored these 50 studies how they categorize it and as you can see we've suddenly moved from four percent than this very high causal we have identified a causal effect to 20 so five times as many are certain that they've actually identified a causal effect and as you can see they are more certain about causality than than the outside experts okay so who who knows maybe the authors are actually right this is also to some degree subjective assessment but when we then see and this is really the key point when we then see the media studies suddenly 44 of all media sort of like newspaper articles newspaper articles about these 50 papers almost half of the used language is caused by is due to more coffee drinking leads to higher whatever or eating more chocolate leads to higher whatever it could be they have identified they they claim

that they sort of portrayed as a causal effect
even though the original studies actually didn't try to do it themselves or certainly shouldn't have done it themselves so it's a fundamental challenge that we see both in scientific studies but in particular in the media stories in newspaper articles about scientific studies that they tend to oversell how certain we can be about the conclusion in this in this research and and this is then also why i think it's a fundamentally important competence to have to be able to read a newspaper article and say well not just be skeptical we can all be skeptical but be skeptical in an informed way where we can say well because the data has been collected in this way i actually can figure out that we should not trust this particular conclusion because for various x y that reason whatever happens in the particular context we cannot be certain of it and we'll see a lot of examples of that both in the next lesson on experiments and also research design in general we'll be getting the tools to become much better at identifying how much we should trust the results of a given study and this is even if we don't know the theory that it's involved we don't have to be field experts there are some sort of fundamental things about research design that we can pay attention to so that's sort of the challenge this is basically just to identify the challenge that we have um and showing that this is a challenging newspaper articles and what we want to get is the tools to be better at this

The Nature of Causation

but i also want to sort of say a few more words about what what does this causation then mean okay we know that it means enhance or impact or leads to or has the consequence so all these other sort of everyday terms we can use but but is it actually the true causal effect and i've sort of already indicated that no i mean this tweet really highlights it or summarizes it causality is not binary we should not think of causality as either being there or not our causal explanation can have various kinds of certainty so it's continuum like we see here it's a continuum of being very uncertain to being very certain about the truth of a causal effect um and and so while we in theory of gravity could say well it's really deterministic and in our field it's very much probabilistic so the impact of incentives on solving simple tasks is fairly high but it's certainly not a one-to-one match so that's a key thing in social sciences we should think of causality as being likely in addition to that we should also think of well not it's not only a question of whether it's sort of a one-to-one deterministic match or whether it's just probabilities but should we maybe often think of multiple causal effects because the word cause might make us think of that there is just one thing one thing causes something else sunshine causes sunburn but as even John Stuart Mill said quite a while ago it's rare that there is a single

antecedent there is rare that there's a single thing that comes before the the consequence there's usually a sum of several things that need to be in place for an effect to happen so often we shouldn't think of one cause yes smoking has a big impact on lung cancer but there are a number of other factors that can either decrease or even completely eliminate the potential impact of smoking even such a simple question as does lightning cause fire actually illustrate this because yes i guess in some sense it does you can see situation where there is a lightning strike and a fire emerges but i mean is the dry ground what's the temperature has it rained are there other random factors that play a role it's not sort of lightning as a single factor that will always sort of be the sole reason for why something is likely to why a fire is might emerge um there are often multiple causes for something to happen an organizational success is not depending on one single factor is depending on a long long range of various different factors and that's also already emphasized it's not just a matter of probabilities it's not just a matter of that that the causes are often coming from multiple places it's also an average causal effect that we usually dealing with in social science so just because there are some humans that don't respond to a cause in the same way as others i mean that's just the way life is and then something will also dive deeper into we still think

that there generally is a causal effect
even though not everyone
benefits from a certain kind of drug as
the medicine slide illustrated the last
time
and then also just it's a very basic
philosophical question of whether causal
effects are even real
because we can't really see them um
well we might be able to see this feel
the sunshine that could sort of create
the sunburn
but it's not like we can really dig down
into the skin
just with our bare eyes and see the
actual causal effect
of how financial incentives influences
human behavior again
we know what a financial incentive is
and we can look at the performance but
there's something in between here that
it is sort of difficult to
completely specify what it is this
causation
and it's something that we'll get back
to in in several other lessons
um to what degree causation is a feature
of the physical world
or just a convenient way to think about
relationships

The Aim of Research

so overall and that's really also the
key message here is
yes correlation does not imply causation
sorry
correlation does not imply causation um
but it's evidence for causation so so we
should sort of be cautious about this
it's often said correlation does not
imply correlation as if
correlation does not imply causation as
if i mean we don't have any evidence for
it well they would imply is really meant
to mean something strong here because
correlation certainly
indicates causation as also written in
theBarrowman

article it doesn't imply causation but
it does waggle its eyebrows suggestively
and gesture effectively will
mouthing look over there so if there is
a correlation it's certainly worth
looking into
maybe we find out that there wasn't any
correlation but but it's
certainly something to to pay attention
to so we shouldn't really
say doesn't imply causation to make it
sound like there's no relation at all
if there is a correlation there's a good
chance that there might be some cause of
the factor associated with it
but the aim is to reduce uncertainty the
aim is to figure out how can more
certain about the conclusions that we
come up with in science
so science operates as a procedure of
uncertainty reduction as we'll hear more
about in lesson 10.
so it's not truth in the sort of
complete capital t kind of way but but
something similar to it that we often
have to settle with that we get to

How to Infer Causation

but the question then still is how can
we be sure that
actually at x actually causes y uh let's
take an example let's assume that out of
a
thousand firms 500 have decided to
increase salary in 2017 and
the other 500 did not firms where salary
was increased they actually had a higher
productivity in 2018. let's say this is
how the correlation looks like
can we be certain that the salary
increase
actually led to this higher productivity
well these apparently the 500 firms in
this hypothetical example just decided
to increase their salary
but but why have these particular
companies increase the salary why
have we sort of why are these have the

these firms selected into the salary increase maybe because they knew they were going to do well in 2018 and they want to be sure to to keep their employees and they could afford to keep their employees so maybe it's actually just because they knew they were going to do well that they decided to preventively raise the salary to keep their key employees maybe it's just better economic conditions in a particular industry um that that sort of have led these two to a higher demand and then could uh facilitate higher productivity or maybe it's not the salary increase maybe it's something else recognition or new leadership that influence behavior these particular 500 firms so maybe it's the new leaders that decided to increase salary that then also decided to do something else so the thing is that with many kinds of situations like these where it really appears to be i mean it would be meaningful if the salary increase led to higher productivity but again we want to be really sure that it was this particular factor and not some other factor that we're not just paying attention to so how can we be really sure it's the randomized control trial it provides what we call a counterfactual that Barrowman talks about in page 11 and 12. so we have the medicine example we have a group of patients some receive the drug and some don't receive the drug in this way we can not only say let's take a hundred people we give them all a headache pill and see how many don't have headache afterwards that could lead to the conclusion that that headache pill is really really effective

because over time almost everyone loses
their headache
but but what we want to do is to compare
in a given period of time
50 people that get the headache pill and 50
people that don't we want the
counterfactual
we want to know what would have happened
to group b
if group b had received group a's
treatment
um and this is then what we then want to
look into with the randomized control
trial
we want to find out how the randomized
control trial how an experiment
can give of this almost certainty that
it's the headache pill it's the
intervention it's the salary increase or
whatever the example is
that led to the causal effect so how is
it that we should design
such a trial in order to be sure about
causality

Summary and Next Lesson

so that's the aim of of of the next
session
um is to get this understanding of
randomized control trials
the aim of this one was to say look
let's get a good sense of the concept of
correlation, causation
what are some of the challenges of why
sometimes something might appear to be a
causally related
why might it because there are other
underlying factors or because of
selection effects
and and we also then went through well
what does it mean
for something to um what does it mean to
say that something is a cause and effect
should be think of as single or
multi-causes
um and and and
and in social science we should sort of
settle with probabilistic causes that

have an average effect
sort of thing some general insights into
what it means when we're looking for
causality and
this then we're getting some insight
into how to actually do that
in the next videos and sessions

<https://youtu.be/TY1xoPon42U>

we will now continue and focus on
experiments
but while we in the former session focus
on lab experiments to illustrate
the nature of a randomized control trial
we'll now
focus on what we call field experiments
and
and i'll try to explain what these are
why we
run them so what are the advantages
and disadvantages of running a field
experiment compared to a lab setting

Synopsis of RCTs
so again just sort of to reiterate the
main points
the randomized control trial is a
particularly
strong and useful way of trying to get
to the
causal nature of a possible relation
between two
two things so because we
randomize participants into two
more or less identical groups and make
sure the environment around the
experiment both in the control group and
the treatment group is as identical as
possible
and we only change one thing then we can
be quite confident
that we have established whether
something works let's say it's a
headache pill
if we gave the 100 give the headache
pill to all hundred people
and and let's say they all had their

headache removed well
we like the counterfactual we lack
insight into well what would have
happened
if some of these people did not receive
a headache pill and that's what
randomization is for that's why we
carry out an RCT but we also have to

Internal and External Validity

remember as
we heard before that causality should
not be considered as a binary thing
and the validity of an experiment is not
a binary
thing it's not yes or no true or false
it's more of a continuum
and this relates both to the internal
and external validity
internal validity being what is the
truthfulness of what is the validity of
this particular
study in this particular context um so
of the Ariely's study with bionicles
how certain can be that we can trust
these particular results with these
particular groups of people in this
particular setting etc
while external validity is then saying
well
we are not only interested in figuring
out
whether this one particular study and
with
but it's all it's with its particular
setup
is true we are actually also interested
to know can we learn something about
the world more broadly um so we
we focus quite a bit on that on friends
used Ariely's bionicle
study to make us think about well
what's the strengths of the external
validity
to what degree do we think we can say
well it happened to have been run on
male harvard students
do we think that this also applies to

the population more in general
the study relied on a very very simple
task of building bionicles
do we think it will also be meaningful
in in other settings where we do other
tasks
um so we we
the aim was to really get a sense of
that we should not say a study has good
or bad external validity we should be
more specific
in terms of thinking about interaction
between selection and treatment
interaction between setting and
treatment
it's run quite a few years ago would the
results still apply
into the social world we see today etc
so these were the
internal and external validity
trade-offs that we saw in
lab experiments and
and and and and it was clear that
especially external validity
was was challenged i want to sort of
give a little tweak to the concept of
external validity because external
validity is a very broad concept and
while ecological validity that some
people also sometimes use
is is more literally whether whatever
study you've done it also works in the
real world and
and here we have an example i've
mentioned my studies before
we actually have a study that shows that
if you run
studies that you usually run in a lab
with lab mice
and then put these mice out into the
natural habitat it turns out that some
of the results we get are actually
different
so it matters whether we have people in
the lab or whether we have people in the
real world or
here it was mice we had in the lab or
the real world so that's what ecological

is saying is how well are the findings
applicable
to the real world and everyday life so
this is a subset of external validity
but external validity is completely
broad it can be generalized to other
settings
another lab in another country other
kind of participants
this is whether the bionicle study would
also make sense
in an organizational context for
instance
and the point being that you had to
build bionicle that might be dismantled
right in front of you
for very small monetary rewards does
that really match
what we think happens in organizations
do they really carry out such small and
simple tasks
are things really dismantled in this way
so that's the that's the challenge of
lab experiments and that's why we
um want to focus on well well how can we
then get
stuff how can we get these studies out
into the
real world and that's really the promise
of field experiments
because they are run in the field so
sort of to to to highlight the point and
put it slightly differently
so that even if a lab experiment
is as well done as well as possible with
as high internal validity and external
validity as we can sort of imagine
um it's still often generally based on
student samples it
still will take place in artificial
environment people know they're part of
an experiment when you you
sign up to be part of something you you
show up
and you get paid usually these
experiments are often also fairly brief
they might last a couple of hours if
they're really long they might last half

a day
um so what if we want to study real
world situations over a long period of
time and
really involving real stakes where like
whether you get fired or promoted or not
as an employee in a company for instance
rather than difference between
getting 2.40 or 2.1 dollars for building
a bionicle we we want to get out into

Field Experiments and Examples
the real world and that's
what field experiments do we still test
claims of causal relationships we still
randomly assign subjects it's still a
randomized control trial
but it's distinguishing characteristic
of field experiment is that conducted in
real world setting
and often unobtrusively meaning that
participants might not know they're part
of an experiment i'll show a couple of
examples of that
here in the following slides so but the
key thing is
they're done in the field i mean the
name sort of gives it away
um just to give an example of it so
a Ariely's bionicle study was trying to
figure out
whether it's important to see the fruits
of your labor and whether that makes you
more productive that was a
Ariely's sort of setup that's his claim
that's what he wants to investigate
and the argument is that even sort of
such a silly situation where your
bionicle is dismantling right in front
of you
um this is sort of counterproductive but
it's and and and it's one thing that
it's done on males only
that are really good in principle sort
of have dealt with that
but can we actually in a meaningful way
translate that to into a proper
organizational real world setting

well that's then what adam grant is talking about here and and the study that i'm referring to here on the right is well we could answer the same research question so roughly the same research question by saying i want to investigate how important meaningful work experiences are and so here is an experiment done on a group of employees into a telemarketing company that were to call people in order to raise funds for for a good cause of sorts and and what they've done is they've created four different conditions an experiment doesn't have to be only two or three conditions it can be as many as you want almost you're still sort of comparing across these conditions so here we had a control group here we had a group where the employees at the telemarketing company they they sort of heard about good stories that happened when they had actually raised funds for the cause that they were arguing for this did not really make a big difference here in this condition they actually heard about life-changing stories let's say they managed to raise funds for someone's education and they heard about the story of this person actually getting funds for education and this was a big deal for this person of course and this had an impact on how much money was raised this is what the bars here indicate how much money is raised by these employees on average yet meeting the recipients so actually meeting someone who got money uh based on the work done by these

telemarketing employees
really had a substantial impact plus 400%
percent
in in in performance this really had a
substantial impact
so this is just to show a very simple
example the research question was
whether it's important to see the fruits
of your labor
well this is basically what adam grant
did in the study here
he compared to telemarketing employees
that were either just sort of
in abstract learning about how important
their work was maybe they
actually heard about specific story
about any person who benefited from it
or they really met the recipients and
apparently this had a substantial effect
um so now we've investigated in a real
world context an actual organization
then we're divided into different where
the employees were divided into
different conditions
um and so the external validity is
clearly a lot higher i mean still
it might only work for telemarketing
companies or maybe it only works in
i assume this was united states and
there can still be other limits to it
but we've taken it outside the lab so
the ecological validity is clearly a lot
higher
um in our our sort of
trust in whether we can generalize this
should be a lot stronger
and just to give you a couple of more
examples to highlight why we sometimes
need a field experiment to study
something so this is a very
well-known economic study big question is does
it actually
matter to to be sort of to learn about
management practices
so there were a group of researchers who
went to india
and did a field experiment on a large
indian textile firms

so they provided free consulting on management practices to randomly chosen treatment plans so i would say there was a company textile companies who had various manufacturing plants where they where they produced these this is this textile stuff they produced and some of these plants which were selected and they got free consulting so the manager of these these manufacturing companies they got free consulting and then the aim was to compare the performance of those sector got the treatment with those that didn't get the treatment to see well did performance increase and actually um productivity increased by 17% in the first year and which is quite i mean that that's really a lot so these facilities were quite big so this was really really a big impact but of course the question of this management matter in principle you could have created an artificial environment and taught some participants in the lab some management practices and they maybe had to create an artificial company and see who sort of manages the best it's of course far far more convincing to be able to go to two companies and and then not just sort of do an artificial thing but really do it on the real thing still in terms of external validity who knows if this translates the transfers to denmark united states or china we could sort of have a discussion on that but it's it's certainly um it's certainly a much higher validity than any lab study could have and actually the same group of authors actually followed up

on these textile companies 10 years later so they revisited the plants in order to find out well do the effect actually still hold here 10 years later and as i show here there was still a large and significant gap in practices and even at large there was even a significant performance gap between treatment and control plants so even 10 years later um they could still see the effects of the management practices they helped to implement a decade earlier so again this is then showing not only does it have a short-term effect but we can have a field experiment where we can show that it actually also has a very long-term effect logistically this is of course really difficult to pull off this research because it literally takes 10 years to to run but it ends up being a very convincing and compelling case for the importance of management practices

Is this a RCT?

okay i suggest now that you pause the video and and read the um the example given here it's taken from from the textbook chapter on experiments so try to read this and consider is this a proper randomized controlled trial and what kind of experiment is this so if you pause the video you have a chance to read it for yourself i'll just continue here so key things to pay attention to is here okay we it's it's it's experimental participants are sales people working for a north american insurance company so these are actual employees in an actual company and they were invited to participate in a study and they were only you had to be able to commit yourself to four weekly training

sessions etc so
there was some kind of selection going
on here and 60 individuals volunteered
but then they were assigned randomly to
experimental and control groups
so it's a randomized if there's a
randomization going on maybe the
original 60 people
are not typical sales people so maybe
there are sort of particular individuals
who have the time and the
motivation to participate in some kind
of
management in some kind of training
program
but they were randomly assigned to one
or the other okay then
members of this the experimental group
received training and self-management
over several weeks
and the control group did not receive it
until much later in the future
er effect a year later so that's quite a
bit later but here we have an
opportunity to then compare
the performance of people that got the
experimental treatment
with the control group that have not
gotten the experimental treatment yet
and they assessed effectiveness at 3 6 9
12 months and it turns out the
hypothesis was confirmed the
experimental group performed much better
than the control group
at each point on the various
effectiveness criteria employed
um furthermore after 12 months when the
control group went through the
the training program the differences
disappeared and they were sort of at the
same level of performance
so what's the takeaway here then we can
do
a randomized control trial in the field
we can have a randomized setup
we can sort of try to make a
manipulation that is sort of
quite clear either you get the

management program or not
but you could also speculate well how
controlled is this environment how
certain can we be
that people from the management so those
that were in the experimental condition
did not talk to people that were in the
control group we
this is difficult to control we can't
sort of um
at least not in in most countries and
companies you
can't force employees not to talk to
certain other employees and even if you
did that that would sort of
create a weird awkward situation
so we cannot be 100% certain that there
wasn't some kind
of communication so we could actually
imagine that the insights from this
management
course sorry this training program was
quite
easy to convey to others and this could
have led to a situation where those that
did not participate in it yet
heard about it from those that did
participate and and
and then they all sort of improved just
as much so
if the result had been a null result if
they didn't find any significant
differences
let's assume that let's assume they did
not they were not able to find
any significant difference in the in the
experimental treatment and the control
treatment
well this could have been because the
sort of information was leaking between
these two different groups
that's um that's a challenge you have in
these field experiments so this way this
is a nice example of
the quality and the strength of a field
experiment you are actually finding out
whether this works for this particular
company but

but there are risks involved in terms of not being 100% sure about the internal validity

Internal Validity Compromises (Face mask)

another example that we also had the last time was the weather face mask protects against COVID the danish study that i talked about last time and just want to point out that's the field experiment because it's run in the field it's people that live their everyday life that that you either wear masks or don't wear masks so it's an experiment that is in the field and that's why we call it a field experiment we could question the internal validity because actually they weren't investigating whether face mask protects they were sort of more indirectly investigating whether informing about face mask and sending out face mask has an effect um as we discussed the last time but it's it's it's another example of how it's more difficult to control this if we had put people in a lab and just sort of hypothetically had had a group of 50 people where they were all wearing masks and another group of 50 people that didn't wear masks and there was an infected person in both settings then we could have investigated whether they got the disease or not this would not have been we would not have gotten ethical approval for running such a study of course um but in principle the internal validity of that study that sort of that mass study would have been better the internal validity would

have been better
but we want to investigate how people
behave in everyday life i mean it's
difficult to put people in a group
where they might realize they were sort
of in a situation where
COVID plays a role and then they might
change and adapt their behavior to that
here we we had people as they were
living their lives

Is This a Field Experiment? (Super Bowl)
an example i actually used last year as
you can see the tweet is from february
2019 so this is not
Covid-related um you can pause the video
if you want to sort of read
the essence of the study as it explained
here
what it's being shown here is that
influenza mortality in cities that
participate in the
super bowl so that's american football
in cities that participate in the super
bowl is higher
so what they're able to show is based on
data from 1974-2009
they are comparing cities that end up
having
their team so to speak in the super bowl
and then they're looking at
how many the flu death do we actually
experience
right after that super bowl is played in
in late january or
february compared to cities that where
the team is not in the super bowl
and it turns out that there seems to be
an 80 percent increase in the flu death
for the population over 65.
is this in a field experiment no and the
reason is that the researchers have not
allocated cities to participating in the
super bowl or not
this is done sort of by whatever sort of
happens out in the world one team beat
the other team and that's why they ended
up there but it's not

controlled by the researchers it's not
researchers that have flipped the coin
and said
today san francisco gets there and new
york doesn't
so it's still an interesting study and
we'll get back to this example later
um by the way and at the end of
uh lesson three but it's not a field um
experiment because it's not an
experiment it happens in the field but
it's not a randomized control trial

Constant Exposure to Field Experiments (NYT headline ex)

so a point i also want to emphasize is
that we constantly expose the field
experiments even
that we don't know it i told you how
amazon and google and ethics etc that
run thousands of experiments every year
and and you're also part of those you
whenever you sign up to
you you sort of accept the terms and
conditions
of these companies and of course you
read those carefully right um
you are accepting to be part of
experiments and here where it's
being written about the new york times
they often in the morning will test
different headlines so readers of new
york times
will be ex randomly selected so the
readers some readers from new york times
will be randomly allocated to different
groups
and for this particular studies the
question the headline might have been
which businesses should reopen first
your favorite restaurant might be a
super spreader
is it safer to visit a coffee shop or
the gym so basically
the ones in charge of writing headlines
they didn't really know
which of these three might be the best
and generate the
most attention you can yourself think

about one
do you know which one would be the
winner
it turned out to be number three but
this is a good example of a newspaper
that realizes we can try to guess which
headline is the best
but why not run it expose let's say
a thousand people early on in the
morning for this 333 in one et cetera
and
333 in the second and the third group
let's see who clicks on the link the
most
and if a certain headline functions the
best well by all means why don't we just
use that headline
for all the articles the rest of the day
this wasn't just the one
time thing this is something the new
york times does continuously
and i'm sure if danish newspapers are
clever they should do something similar
i don't know if they do
but um it's fairly easy in these
online settings to do this that's a
clear advantage of the online settings
so

Pros and Cons of Field Experiments
so just to summarize some of the
perspectives here we've tried to compare
lab versus field experiment i've tried
to illustrate
why and how we can run these field
experiments but also then
the pros and cons of this so it's clear
that it's more real behavior it's not as
artificial setting if you're in a field
experiment
the ecological validity is higher but
the examples then
of the sales insurance company also
illustrate you might have less control
over variables
you can't be 100% sure that the for
instance participants in the experiments
aren't

communicating with each other or that something else might be going on um so the randomization the clean randomization is might could be a little bit more difficult because you don't know if people from different groups are communicating with each other but it could certainly be a more relevant sample it's not just students that study sales but these are actual people that have to do the sales as an insurance company and and you really want to improve their performance so that you have a certainly a relevant you can have a more relevant sample but it's also much more resource demanding as you can see for instance the the insurance company study in total took what is it 15 months or something at least the the india field experiment on management whether that matters that in the end took at least a couple of years for the first round then another decade for the second round these might be extreme examples but but still you have to convince the company of running this and you have to set it up and it's just much much more resource and time consuming than doing it in the lab who who sort of are going about their their life their work life so they embed it in real life meaning the stakes might also be higher for the indian textile companies it wasn't just participating in a lab experiment and then hopefully being the best this was about survival this was about making a profit this was about which of the textile companies might sort of be profitable in the long term um so this is just sort of this is not

an exhaustive list this is just to illustrate
that there are trade-offs a field experiment is not necessarily better it depends on the research question it depends on what we prioritize internal or external validity
here i've just put in a question from a former re-exam set in 2018
um where one was explicitly asked um to compare two studies of that sort of addressed a similar question one was the lab study one was a field study so so just to illustrate that this has emerged in how this could emerge in an exam setting

Survey Experiments

finally i just want to say that we also do have other kinds of experiments that are not sort of clearly a lab or a field experiment we could for instance do a survey experiment which are quite easy to pull off i know many of you in your bachelor project on your master thesis will do a survey study but what is being done here is you can imagine that you want to ask about in this case is um something about fairness and earnings and the original question was a 50 year old man with vocational training is working as a programmer and earns 1200 euros a month is this fair or not well we could make it a 50 year old woman or a 40 year old man with or without education or we could change the the the occupation we can change the question in many different ways and then see
well let's say we do it with a 48 year old man
is that would that constitute a substantial difference in terms of how respondents

will assess whether getting 1200 euros a month is fair or not
or more anymore obviously we can change it to a
woman or someone with another education so we can
we can investigate a question not by asking everyone the same survey question in order to find out what people think about a particular scenario but we can ask them different kinds of scenarios in order to find out
what really matters so we we would be interested in vaccines and how willing people are to take a vaccine
and then we can ask the questions in different ways in order to really get to a
better and more clear answer than if we just sort of ask it in one way
um so this is also this would also be something that is possible to pull off in a bachelor project in a master thesis
um if one um
if
if this fits the research question i for instance supervised a project where students wanted to sort of investigate uh the role of influencers in social media
and then they compared so what if an influenza that is either very well known or not very well known
promotes a product that is very well known not very well known this sort of simplified version of the project they had
and then they sent out a survey experiment where they manipulated some a few of the features of the survey and got different kinds of responses and then could provide an informed answer on the role of influencers
in terms of being able to promote products we have to keep in mind here with these early experiments that they capture perception they can they capture

what we think
about this situation about the
whether this is fair or not this does
not mean that this is how we would
actually
act maybe we will have different
reactions if we actually meet someone
who fits this categorization
same with the social media study maybe
people will respond differently
differently if they are
on an actual social media site rather
than answering surveys in the artificial
survey experiment setup
it's something we'll return to in in a
later session but just sort of to
emphasize
survey experiment by nature what you get
is how people sort of
take a box which is very different from
a field experiment
where you actually want to see how
behavior changes
but this is just to highlight that yes i
make a main contrast between lab and
field experiments but we could actually
imagine
other kinds of other kinds of
experimental setups

<https://youtu.be/U9GNoIQ0qOw>

hi in this video i'll be presenting
different kinds
of research designs so in a second i'll
explain what a research design actually
is
and then go through three of these four
types of research designs we've already
covered experiments so i won't repeat
that
but i'll be explaining what's sort of
the key characteristics of each of these
research designs
and what are the strengths and the
weaknesses associated with each other
keep in mind here that the learning goal
is not just to be able to say

look this is how we define a cross-sectional design or a case study but to give us the tools to assess actual research being done that we read about in a newspaper or a research article or a textbook and to assess the research critically and to consider whether this particular research design can actually answer that question that was asked this is what you also saw in lesson one the example from 2018 exam case the question five with Brenda and we'll be looking at that example again in the study cafe in the next lesson this sort of this example quite nicely illustrates the kind of learning that I want us to draw from this the research design

Research Design vs Research Methods concept can be contrasted with research methods so research methods would for instance be to engage in qualitative interviews and rely on simply structured interviews to collect data and then you also make some decisions on how to analyze and code data afterwards for instance or it could be decision to engage in survey-based research and rely on regression analysis to to to make sense of the data these are sort of more specific decisions of the particular elements of the research that you engage in research design is an overall frame so it's the overall decision for instance to collect data twice both in the beginning of your project and at the end of your project and in order to compare whatever results you get or it can be the decision to engage in a randomized controlled trial and allocate

um a treatment to one group of people
and not do that to another group of
people
or it's the decision to look at multiple
cases in order to make sense of your
research question rather than just look
at one case
so these are some overall design issues
that one might engage in
so it's not whether to include this
variable or this case or
or sort of similar challenge it's
again it's on this more overall level
and
a study's research design can then tell
us whether the evidence that one has
collected
is then actually able to help us answer
the question that is being posed as
unambiguously as possible
so we'll see in the following examples
and the rest of the session
illustrations of this and how a certain
kind of study for instance a
cross-sectional study
might not always be able to answer the
question that the researcher has posed
so this is the research design angle
that we are looking at here
and and note also as stated here at the
bottom
the choice and quality of research
design very much depends on the research
question
again i'll get back to that

Overview of Research Designs
and again as mentioned it's the three
these three kinds of research designs
that we'll be getting into
and the learning again is to be able to
say
not just what research design is in play
in a given situation so
it's not just about identifying a
research design but also being able to
assess the strengths and weaknesses
of a given research design and for

instance if
in a particular case actually warrants
if the conclusion is warranted based on
the research design as
in play okay so let's get to it

Cross-sectional
cross-sectional research design the
analogy is all from this
snapshot idea you get um
you get insights just like when you take
a photograph you get a snapshot of
what's going on right here right now
so you would have more than one data
point you would for instance have data
on 100 people
at a single point in time and you will
know at a single point in time what
their height is and what their weight is
or or you would have insights on how
motivated they are and what dedication
is and what their age is and you want to
find out well
what explained productivity it's really
the core of many quantitative methods
courses
so the aim is to explain variation the
aim is to explain that on average
which variables for instance this
motivation education age example
which of these variables are associated
with the outcome
with productivity and we might find out
it's in particular the indication that
one has and motivation that that
is statistically correlated with
the outcome with productivity and what
this then can do
is to establish correlation but as we
heard
in a formal video correlation does not
necessarily imply
causation we might not necessarily know
whether it's motivation that causes
productivity
or whether it's because one is very
productive that one is also very
motivated to engage in a given task it's

difficult to
disentangle what comes first because we
only have data from one point in time
to sort of elaborate a bit of this
example and to highlight what we're
going for
so we would have for instance some kind
of spreadsheet with data
with data on person a and person b and
person c etc
we might know their education and and
motivation variables et cetera and their
salary or productivity
and then we can do some kind of
regression analysis to find out and
on average how does different types of
education for instance
impact salary or productivity whatever
we're going for
we're not really interested in each
individual person
as such we're not really interested in
why a particular person
chose a particular education ended up
with a job that
landed him or her a particular salary
we're interested in the average
variation
that the associations between different
variables
lead to so it's the average course that
we go and go we're not going for
understanding a particular person
but but again when i say cause we should
keep in mind it's it's always
cross-sectional it's it's it's
it's difficult it can often be difficult
to establish that
um it's motivation that leads to
productivity and not the fact as i
mentioned before
the fact that one is very productive
which then leads one to
become very motivated as well

Longitudinal

so this then stands in stark contrast to
what we call the longitudinal design

so in cross-sectional design we would
for instance compare 20 and 40 and 60
year old and we'll
carry our research in the year 2000 and
we'll see
whether each of these groups are more
motivated or more productive or whatever
our issue is
in a longitudinal setup we would study
in this example here the same 20 year
olds in 2000 and then when they were 40
2020 and
when there are 16 2040 we would study
the same participant
and vary at various points in time so
this is really the key thing here is we
have
data over time on the same people we
don't just have data on
some 20 year olds and then in principle
we could collect data on other 40 years
olds
a couple of weeks later this is not what
we mean we have data on the same units
at least twice as highlighted here so we
could for instance have
data on the motivation of employees the
time point one let's say in
january 2021 and then we could look at
how motivated and productive they are
in february 2021. this might enable us
to say
well it's actually a motivation in a
prior time point that explains
productivity at a later point in time so
this improves our chances of
establishing
some kind of causal connection um
because we're going to identify what
came first
in a cross-sectional setup we can so
because it's the snapshot it's a picture
taken at one point in time we can't tell
what came first it's just
it's basically not possible at least not
if we just go by the data that we have
um in a longitudinal setup we would be
able to say well

since we have data from two points in time we can say that something comes before something else otherwise longitudinal research can in many ways be the same as cross-sectional it's just adding an extra time point of course it can be you can collect data from the same people every week for a year or every month for a year or every hour for a day as long as you have multiple um as long as you've collected data from the same whether it's individuals or firms or cases or countries or whatever it is at multiple time points which because this enables you to compare development over time and it will potentially enable you to say what happened in time point one is actually explaining what happened in time point two we'll get a few more examples in that in in this lesson four and this can be done qualitatively it can be done quantitatively by sending out the same survey multiple times but it can also be an individual who actually observes a group of people over a long period of time and this can also constitute longitudinal research then as a as a third one presented here

Case Study

in this video we then have what we call case study design which is of quite fundamentally different nature here we are looking at some kind of bounded situation it could be a unit in an organization it could be an event um it could be a person that we are following or multiple

persons that we are following
and here we're really interested in the
in-depth reason for whatever is
happening so before i said explicitly
in the cross-sectional study we wouldn't
really care about why peter did
something we just look at
how various variables for instance the
motivation and age and education of a
bunch of people relate to
their productivity and we don't really
care about the individual as such
only in the sense that we want to get to
an average result for instance via
regression analysis
while in the case study design we're
really interested in the individual
persons and why is it that peter
um has the particular education
and has this why is it that he's
motivated in the way he is and why does
it lead to a certain kind of
productivity so we don't really care
about the averages of a lot of people
because because we don't have
data on a lot of people um
we we want to focus on the con this
particular context the
unique context we're looking at this
case study design doesn't have just to
look in one case we can look at multiple
cases
um and compare them again we wouldn't
sort of do any kind of statistical
analysis usually
we would in a qualitative sense trying
to compare
differences between different cases
maybe we have a unit in one country and
a unit in another country and want to
find out what the differences are
between these units
case study design can also be
longitudinal we can also follow a case
over time and thus have
multiple time points multiple data
points on the same people over time
but again not with the aim of

understanding some kind of average correlations but understanding why stuff develops over time so it's really how and why something is happening that we're going for typically this could be interviews observations but you can also use surveys or archival data or other kinds of approaches so whether something is cross-sectional design or case study well depends on what you're going for um you might be comparing 6 8 10 12 different kinds of cases but what what answer are you looking for are you looking to try to understand each particular case and what is particular about each case and and then what happened in each situation or are you going for some average variations and and sort of caring less about the particular context so as mentioned here then we can also have single versus multiple cases and um and there are some trade-offs and advantages of each of them if you're doing either single multiple case study you can read this paper that provides more insight into the pros and cons of each approach the more cases you're looking at the less in-depth you can look at a particular case i mean basically think of it as a resource question you have the resources to carry out let's say 10 interviews well you can do you can take 10 cases 10 individuals and interview them all once you can look at five cases and interview them twice over a month or two or you can take one person and interview this person once 10 times or for instance once every week over a 10 week period so it's all about how you allocate your

resources and the more you focus the fewer cases you focus on of course the more in depth you can go with this particular case

Differences

so they're really trying very different things
um the cross-sectional study is doing this snapshot while the longitudinal study is like the video camera that is um is tracking how stuff develops over time while a case study can be longitudinal you can follow a case over a long period of time but you don't necessarily have to i mean you can do a case study where you just go out one day look at what happens in a company and unless you're explicitly aiming for the difference between for instance 10 o'clock in the morning and three o'clock in the afternoon we would usually call that one time point or something you're going out there to collect data once and trying to understand that so it becomes longitudinal the moment there is sort of something you're interested over time that you're interested in in differences over time something happened maybe new employees were hired in the company that you're you're observing or doing interviews at and you're interested in how that changes their responses if we can think of it in this qualitative case study setup

FAQ and a Short Test

i have here in the following slides provided some answers to some comments or some typical questions i'm getting throughout the years i won't

sort of go through it and read it it's
it's you can
read the bullet points yourself and it's
just the
answers to some typical questions that i
would usually get
and it also relates to some of the
points that i've already
talked about and i've also inserted
very sort of simple multiple choice uh
setup where you can look at these
questions and the answers are the next
slide
just to see if you're really getting the
the basic elements of
this of these different designs right

Differences (continued)

so just to jump back to this slide here
in order to assess whether something is
for instance cross-sectional
longitudinal
try to pay attention to how is data
collected is data on the same person's
being collected multiple points in time
so it's not enough that you have some 20
year old and some 40 years old you
actually need
the same people need to collect data on
the same people twice
um so also a survey that sort of goes on
for a week where you get
one response from from each respondent
just because the survey sort of
takes i mean it can be responded to over
a period of
of a week as long as people have only
responded once and and
this is what you're aiming for well then
it is a cross-sectional design
and in a survey you would usually then
only be looking at sort of average
interactions between different variables
as you
will ask people well on a scale from one
to seven what do you think of this or
that
while in a case study you won't just

want to get these very generic answers and generic sort of ratings for instance on a Likert scale but you'll want to ask the person well what is it that you're thinking about this and then why is it that you made this choice and what makes you rate this question in this way so you really try to dive into the details of people thinking we'll get more good insights into sort of further challenges to various kinds of research designs but also get a bunch of examples that makes a lot enables us to get a better sense of what these different designs can do and what their strengths and weaknesses are we'll be getting that to in the live session as well

<https://youtu.be/iVBvrr48kDQ>

in this video i will be addressing some further research design challenges that research might be exposed to so while the research design 2001 text is this classical textbook chapter it provides an overview and some general strengths and weaknesses then this AMJ 2011 text is on a more advanced level and trying to provide insight into some additional and challenges that research design might face so in an exam situation don't expect all of these to come into play but you might have an opportunity to identify that one or some of these challenges are in play in the description of the research design so again this text is written for researchers um and it's trying to outline what are

some typical pitfalls and typical reasons for why a submission of a paper to one of these world's leading scientific journals might be rejected so it's trying to outline some some typical um flaws some typical things to pay attention to and i'll go through all each of these in in the coming slides but also note what they write here that there are no hard and fast rules to apply here so matching research design to research question is as much as art as a science so these rules are not completely clear cut sometimes they can be ambiguity but it's important to know about the um these criteria and these potential pitfalls because it can help us develop stronger research designs and it can help us identify a given study that we read about whether it's a newspaper or a textbook or a scientific article it might help us say look they actually overlook this or due to this particular reason we should have less trust in the results so let me try to go through each of these elements here

Cross-sectional Data and Change so one of the main elements that i've also addressed in the in the form of video is that if you have cross-sectional data it is difficult to address change so let's say that you have data from 2015 and you have data from thousand companies and you know which companies had an increase in productivity and you know which companies had an increase in salary

you will not based on such a cross-sectional data set where you only have data from one point in time it is difficult to know whether it's productivity that led to an increase in salary or whether the company offered an increase in salary and the expectation of this then led to a later increase in productivity it's difficult to pinpoint the direction of this arrow here when it comes to sunshine and eating ice cream we know that eating more ice cream doesn't lead to more sunshine so we know it's this way around but often in in organizational context and business context we might not really know in advance what the direction of the error what the direction of the causality is in this case if you had insight into the productivity in 15 16 and 17 and simultaneously if you also had insight into salaries you might be able to see that salary increased first salaries increased already in 15 and 16 and then only in at the end of 16 and then beginning of 17-18 productivity began to increase let's say you had data on on a monthly basis in this way you would have longitudinal data you would have data on multiple time points and you would have a stronger foundation to be certain that it's actually that the arrow has a certain direction of course you could also do an experimental design have a thousand companies 500 of them will increase salary 500 will not and then see if that influences productivity there's also some fancy statistical analysis you can do this is not

something we'll be
getting into but this is just one of the
general points and that we often see and
also this if you remember from the
causality correlation video
i talked about this overview of 50 most
shared posts on social media
this is what often happened is that they
will have data on one point in time and
then
end up making a causal argument even
though that there could be many other
reasons for
why we see a correlation

Omitted Variable Bias

next element to look at is what we
call omitted variable bias
this is when we have the situation where
important variable
where important data is not collected
and this could then
lead us into a wrong direction so
um at the bottom here i've outlined the
student bookshelf and future salary sort
of maybe a little bit of a silly example
but it still
gives you an insight we might be in a
situation where we can say
that the more books there are in
a student's home the better the academic
achievement
the student will reach this is a
correlation that is very likely true the
more books there are in their homes over
of a student the more books there were
in their home when this student was was
growing up
in at his parents house the the
better and the higher the academic
achievement and but but this kind
of sort of setup might make you think
well let's just put more books
into the house of um of
families because then more people will
end up have a stronger academic
achievement
and that's probably not the right way to

think of the causality because
it's probably the parents intelligence
or educational attainment
that decides whether there are many
books in the home
and it's also the parents educational
level and the sort of the parent push
towards the students future education
that then
influences academic achievement so yes
parents influence the number of books
at in in the home of families but they
also and probably more importantly in
other way
influences the academic achievement of
students so this is a
example of if you'd only collected data
on number of books on academic
achievement
you won't get to the actual variables
that really are important
one could also imagine an example of
Covid 19 and we could look at different
countries and then see how many people
die of Covid 19 and then we could sort
of
um that into a conclusion of well
certain countries are better at
preventing
Covid 19 death than others if we don't
consider that different countries have
different demographics so different
countries have more
less number of old people and the fact
that all the people are more at risk of
dying of Covid19.
well then we risk making the wrong
conclusion that certain countries are
better at preventing Covid 19
while the main reason actually could
just be that some countries or some
areas simply have
fewer people of old age which is then
the reason why they have
less people dying of Covid 19. so
i could sort of continue to make lots of
more examples the point is that
we should be cautious about trusting

any relation between x and y until we've made sure that there isn't any other variable out there that is correlated to this and it is actually playing the key role in the example of the elementary school study and how different points of time they had an influence on how well they did the test well if it turns out that the students that were really really good also got the test early in the day well then you've omitted an important variable which then influences and skews you and destroys your conclusion so again it's always important to try to have data on all the relevant variables and this is then also again the advantage of an experimental setup because you can control that there isn't any sort of random or systematic variation sorry there isn't any systematic variation going on that we can't control so omitted variable bias we often

Inappropriate Samples

also see inappropriate samples being used so let's say you're interested in how top management engages in decision making and then one might run a study for instance a laboratory experimental study where you used regular students to try to dive into this issue well that might not help understand how top management makes decisions because these students are not by definition yet top management so maybe you could use MBA students but actually you really want data from CEOs to really make sure that whatever you end up finding out really matters for how top management makes decisions

um

one could also imagine a study on whether face masks actually helped prevent Covid19 and if this study has been done on on nurses and doctors and then shows how well they are able to use mass to prevent getting Covid 19 well that might not necessarily translate to the rest of the population who doesn't have the same training in using masks um so again one would have to make sure that you don't just sort of transfer insights from how specialists are able to use mass to the everyday population so make sure that when one assesses the conclusions of a research study or looks at the data and ensures that whoever is sort of the foundation in the data set what the data that you've collected and the people you've collected data on that they're actually also relevant for the conclusion that you're trying to draw another example of this is a slightly different example is is google and google the company has realized that general test scores what they call sat scores in the united states or grade point averages don't really predict how well google employees perform at the job so it's just not really particularly helpful when google looks at all the let's say a thousand people that hire every year um the grade point average of these people from their college degree does not really help explain who performs well at google and then one could think well should a typical danish firm then also just ignore grade point averages well google only

hires employees with very very high grade point averages basically if you are from a danish institution you more or less need an average of almost 12 in order to get a good chance of getting a job at google so there isn't really any variation in jp grade point average to take departure so of course if there's no variation in the variable that we're interested in this variable is not going to help you explain how well people perform similarly if we look at the height of nba basketball players in the united states they don't really predict scoring very well this does not mean that height doesn't matter of course height matters it just means that almost everyone that gets into the nba are already very tall which means that this factor is not going to play a big role so um typical danish firms should probably not hire like google because the variation in the kind of people that you get in and the grade point averages it's much much bigger so one should not necessarily look at google and amazon and apple etc to see how they hire people and just copy their their hiring approaches because they might not really translate very well to other kind of settings where you get other kinds of applicants this

Selecting on Dependent Variable is also related to this issue of selecting on a dependent variable that i already mentioned once so let's say we're interested in how successful entrepreneurs they do well and we end up only studying successful entrepreneurs

well we might not really find out that unsuccessful entrepreneurs might do the exact same thing maybe they try to engage in most of the same practices but just fail for various reasons whether it's luck or or a factor that we have that we have omitted so if we want to explain how a certain group does well entrepreneurs or basketball players or ceos or businesses or whatever we're thinking of we should not only look at those and actually do well because we risk missing some really key information for instance that those that don't do well just end up doing the same thing in principle there could be industries where it's pure luck that decides whether you succeed or not and um and you'll only find out about that if you look at both those that do well and those that don't do well so always pay attention to in a study if they say you should do this or you should do that well who have they looked at in order to reach that particular conclusion you can also imagine a study that says look at all these companies that successfully manage the Covid 19 crisis this is how they survived well again look at those that might have tried the same thing and failed and so don't just select on the dependent variable don't just select on the outcome that you're actually interested in but make sure that you have variation in that outcome

Careful Construct Creation

then the article also talks about the construct creation or what's the sort of

the relation between the data you collected and the actual theory so and again maybe insightful to refer to the face mask example that i've given before if you claim to have measured the effectiveness of face mask well it might not be ideal it's not really ideal only to actually get data on whether people received face mask in the mail and an encouragement to use them that's not directly measuring the effectiveness of face mask that's measuring the effectiveness of encouraging people to use face mask so the theory should not be on the effectiveness of face masks the theory should be on the effectiveness of encouraging people to use face mask and and the same thing with all the range of other elements that i'm mentioning here always pay attention to if you claim to have measured employee performance well what is actually being measured here is the manager's subjective assessment of it is it some more objective criteria that is being followed and saying with firm performance it can often be measured in many different ways so one wants to be pay really attention to that the concepts you use in your theory and the words that you use in your theoretical framework also match the actual data that you collected

Common Method Variance/Bias and then there is one mention that is probably a little bit more um complicated to get your head around something called common method variance common method bias it has both of these

names and the point is that you should be careful for instance in a survey setup if your independent independent variable stem from the same source let's provide an example let's say you want to investigate what employee characteristics leads to high job performance we could have a setup where we send a survey to managers that are to rate both the characteristics of the employees and the performance of the employees but this then could create this common method bias because it could be that the manager's particular thoughts on useful characteristics maybe a manager in particular likes a certain gender or certain education or a certain attitude and this then also leads the same manager to assess the performance of these employees in a particular world so it's actually just the manager's liking of a certain kind of quality that also leads them to rate the performance higher even though the performance in actuality is not necessarily higher so we have the risk of a common method bias because the data that we're getting on both the independent and the dependent variable is from the same source we risk that it's systematically skewed by the same source assessment of of both of these elements so if the source is biased the biases influences both the independent and the dependent variable in the same systematic way so you might end up with something that is very significant and looks very substantial but it's just an artifact of of the way that data is collected so

just then briefly put it's much better
to have an
independent source on each of these so
ideally you would have
one for instance one manager or some
range of people assessing the
characteristics of individuals
and then another range of people or
other managers
assessing the performance of these
employees because then you don't have
this
systematic overlap in where your
information is coming from so
ideally you have multiple sources to get
your data from
it just leads to it stronger
just leads to a stronger data set a
stronger design
and a stronger ability to get to the
conclusion that you actually want to get
to

Summary of Research Designs

so these two videos have tried to give
an overview of overall research designs
and strengths and weaknesses of them and
also some
particular more sophisticated elements
that the amj article goes into
this AMJ article also goes into topics
that you will be
required to get a have an understanding
of
to already have an understanding of when
you are at the master's level so that's
why i'm also already pointing to them
here and here the last line i've just
given some very general and some very
rough rules of
thumbs of advantages and disadvantages
of the various kinds of research designs
these are really
just very rough it's not always the case
that a cross-sectional
design will have a strong external
validity and etc etc it's not always the
case an experiment will have a weak

external
validity but these are some sort of
general rough characteristics
that can help one make sense of
different research designs
in the next session and in the live
session they will then be focused on
quite a few different examples of
different research designs and how they
can complement each other
in order to get a sense of what do they
do what are they and and how can we
assess the quality of these so that will
be
the focus in the in the next session

https://youtu.be/kH_BXsAmEt8

they present an exercise and some
remarks on two of the exam questions in
the 2009 re-exam the question set
first an exercise inspired by current
events so again as before as in the
study cafe I suggests that you'll read
the information here there is some
information on what happened in two
different regions in Italy and down here
on the x-axis
yes time and on the y-axis we have
number of Corona cases and the graph
indicates what happens in these two
different cities so I suggest you spent
a couple of minutes thinking about
randomized control trial quasi
experiment something else what's going
on here so after you've had the
opportunity to think about this for a
little while having posted I here
present some thoughts on it so what do
we see we see that two different cities
areas in Italy made different decisions
in terms of how to handle the
coronavirus Lodi engaged in social
distancing Bergamo did not and we see a
clear difference in terms of how the
number of cases has developed over time
is it a randomized control trial well
the data does show clear difference

between the cities it's argued or the way it's portrayed is that there's sort of one variable that is different we don't know if these two cities took other choices we don't know if they did anything else maybe it's not only social distancing so so yes the cities are different but maybe also the people the cities are different maybe better if a government just has more old people maybe low he doesn't have maybe it just has far less there could be different reasons in place so we can't really call it a quasi experiment because cities have self-select into these different treatments but it's still the data set that strongly indicates that there could be something going on about this social distancing the value of that so it's certainly a longitudinal data set that shapes development over time and it's potentially shaped by this one key difference calling it a randomized control trial or quasi experiment we can't really because we don't know how many other things have been in play here okay next up is then question 1 from the 2009 re-exam and again I will suggest that you spend some time on trying to read this case read the questions as outlined up here there are two different studies and try to think about how you would answer these questions and then in a minute I'll provide some thoughts or so if we when you've an opportunity to call it think about it we can move on you to the next slide where if for one I've just highlighted some key elements to pay attention to I have some more remarks on the next page but just want to highlight a couple of the key things so here in study one its made clear that the participant decides when to show up based on their own assessment of their sleep and some students show up having slept well the first time others lip roll it the first time they showed up so there is not a

randomized allocation of who sleeps a lot and it doesn't sleep a lot there is self-selection into when you show up so the sleep number of hours you've slept hasn't been manipulated and so some one thing to worry about is well what might be the reason for why students sleep poorly maybe they were stressed and worried and have stuff to think about which might cause and sleep poorly which might then again cause them to perform poorly on whatever tasks they were supposed to carry out so we don't know if it's an effect of the sleep because we hadn't manipulated it as we would like to do in a randomized control trial so again there is this self-selection into the two different treatments or whatever we should sort of sort of call them so there's an on average variation there is focus on comparing across two different kinds of groups there was no longitudinal data such we're not comparing before and after we're just comparing two different sets of students who have slept either a lot or a little we we don't not a measurement of what happened before something after something so it's cross-sectional design and let's sort of the overall assessment of a word that we want to call it an RTC we could make some further remarks on the internal validity and I've already mentioned this that we don't know why they sleep poorly it might not be an effect of sleep it might be an effect of something else but there's also something about the measurement of sleep where we can see in contrast to study 2 the measurement of how well people have slept is not as strong do we always know if you slept poorly or well I guess we have an intuition and some might have smart watches that captures it and then we haven't sort of taken this into account maybe some of the people have a very good measurement and and maybe others won't

so on top of not really quite being an RTC on top of there might be an element of people sleeping poorly because of something else meaning that there's something else potentially the stress or the worry or whatever it is that is actually causing variation on some of that we also have a poor measurement of the contract of sleep that we interested in and again on the next slide you'll see more I'll just move on here just to sort of let you know there is something more coming here about some of the remarks I'm making here in terms of external validity it's it's sort of unusual in compared to the examples we've seen so far that they've actually done this study and in four different labs around home at around the world but at least in Europe and North America so that's stronger ability to generalize from the sample they've relied on to a broader context it's not just so one particular narrow geographical area but they were still this interaction of of setting and treatment so it's a particular kind of task and dealing with conflicts in the lab is certainly quite different than dealing with conflicts in the workplace in an actual organization so there is this interaction of setting a treatment and and there is also interaction selection and treatment of the kind of people that selected into this in part because of the areas that come from but also who actually does take part in studies at universities they might also be particular so we have this couple of reasons for why the external validity should be sort of limited and then since it was happening in a lab and as usual there's also a weak ecological for the disease you don't have a lot of information about the task they had to do so difficult for us to say a lot okay so study two um here we have an allocation of half the participant into one group and they

sleep less and the others sleep more
don't have explicit information about a
random allocation but we guess we could
assume that just point it out but but at
least there is a clear treatment and
certainly stronger internal validity
then study one

I guess one could say why were they
woken up during the night why couldn't
they just sleep less and then we woken
up after six and a half hours so there
are different ways of implementing the
treatment sleeping lists there's
actually variance just like Nelson would
say there are different ways of
measuring if one I slept less when you
wake them up every three hours to just
make them sleep less in the end just
another example of this heterogeneity
and external validity we have the same
threats as above and we don't be only
even though you had one lab so you don't
have that variance across the lab and
sleeping in a lab is certainly also very
unusual there might be some kind of
automatic that's also in play here but
your overall study has stronger internal
validity because you've had this
allocation of people into different
treatments it's mentioned we have
overview of some of the comments I've
made here that you can also peruse and
get a handle of what are the key
elements to pay attention to okay
another another question so please read
this think about how you would answer
this and I'll provide some remarks
afterwards which are also highlight
slide okay so we have this quote from
the physicists Heisenberg and as you
will hopefully notice this is the quote
that was also shown in class not just
this year but was actually also shown in
class last year it wasn't a surprise for
them to to see it and then these are just some of the remarks that you
could offer here and please rely on the
curriculum to expand upon and explain
what the statement means and we could

for instance say there was something about this we don't observe nature itself but nature exposed so apparently there is something that can be exposed there is something out there to be revealed it's not we don't reveal nature by itself but we could start whatever sort of way you could finish the sentence so something to be exposed so sounds very much like those positivism as Guba says although a real world riven by real natural causes exists it's impossible for humans truly to perceive it with their imperfect sensory and elected mechanisms does sound very much like the groats right so it's to me logically speaking as Guba says it's some sort of modified objectivity we cannot obtain truth object with objectivity in some kind of absolute sense and then an answer one would then expect you to expand upon this post positivist position and why it's not the other positions why isn't that positivism or constructivism fits but but this is sort of the key elements to get to and some of the key perspective from in terms of the isms to invoke we could also rely on Kuhn and say well this phrase the method of questioning Kuhn then talked about how we have certain paradigmatic ways of trying to understand the world within different paradigms we have beliefs values techniques etc we have these four different elements of the paradigm and and so we don't observe nature as it is in itself because we have paradigmatic glasses also to speak we we have some expectations and education and experience that shapes whatever we see out there interpret so one word one wants to expand this a bit more good answer what we would talk a little bit more about what a paradigm is and how and for instance the throat about how we literally don't see the world as it is in itself but good

interpretation always ends where we're seeing interpretation begins when when seeing ends so you would you could also emphasize that quote finally we could also draw upon Nelson and how he says there are some genius variables that fuzzy boundaries don't have direct access to nature as it is we have different ways of trying to measure it whether it's makes you understood as unemployment or motivation or research and development in your firm performance so sleep or whatever it is so the number on the assumptions behind its construction for instance the number of how we measure performance whatever it is comes to between us what the phenomenon is rather than as a useful if limited measurement of that phenomenon so it's sort of the similar point of view about this expose to first time that we don't get the a sense of how the world actually is in itself even though that's maybe what we think we treat it as if we see the world as it is in itself but as he says it's a limited measure there was some these are constructed measure that

<https://youtu.be/62QVoC7QBG4>

released a couple of questions on experiments what is a natural experiment what is a quasi experiments what is an RTC's that I decided do a quick video on this I would like to add though that it's not a key aim of this course to be able to tell for instance the difference between a natural and a koala night experiment the key aim is to be able to say look this these are the characteristics of a true randomized control trial this is proper randomization we have identified a causal effect versus a situation where and then a setup doesn't quite live up to these requirements and thus turns out to be quasi experiment not a quite

a true experiment that's the key goal just to reiterate what is it that an RTC is first of all we are really keen on this randomization we need a randomized allocation to each treatment we start out with a bunch of people whether they're all male Harvard students as in Donna Riley study or Chinese factory floor workers or whatever they are we just want to try to make sure that there are as similar as possible after the randomization we know they're not going to be completely identical people companies are different but we want them to be as identical as possible we could even do some some post tests where we we for instance make sure there was an equal number of men and women age group that have the same age etc okay so we want this randomization to have worked out and we want two groups to be as identical as possible and we want this to to continue to be the case and and we also want them to be in as control and similar environment as possible we we know that if we put people into different classrooms and going to change the teaching styles that while the classrooms might be slightly different but so we're going for this this aim of having no systematic differences between two different settings and we want these very controlled environments because we might want to make sure that the two different groups are similar as possible in a similar environments as possible and only exposed to one particular treatment whether it's how they built Bionicles of whether it's the curtain or whether it's a time of the day where they engage in tests any kind of treatment we could imagine to make sure that there is only this one difference between the groups that we have in mind and and in a quasi experiment what we're then trying to pay attention to is is to say well okay we

realize in the in the elementary school example about cognitive fatigue for instance we realized the location wasn't randomized as such it was people employed at various schools around the country that tried to make sure that that there was classrooms available IT equipment available and that's why sometimes it was for a that had the the early time and sometimes it was the class for PE at the late time but we could then argue that the process was as good as random because kids didn't self select into I want to do the early one today or school teachers didn't self select their classes into certain times of the day it was just as good as random a process and and and again in the quasi experiment if it doesn't happen in the lab or we have complete control over everything we can't be hundred percent sure that there is only this one difference but but hopefully in the end we'll have been able to identify the one true one key factor so in to take the example of the Spanish flu example again we we had that the Spanish flu example when the x-axis have how quickly cities and each circle is a city how quickly cities responded to the crisis and close down schools on the y-axis we have the the impact of the flu had on whatever mortality in the given cities we see a fairly strong correlation here zero points 37 that's far stronger effect than the headache pill has on the reduction of headache question is is this a quasi experiment can we say that there's only this one variable that makes Pittsburgh and New York and st. Louis in Chicago different well actually if we look at the original study we can say no there were some interventions that the city could engage in and there were school closure cancellations of public gatherings isolation and Quarantine so there were actually

different things in play here there wasn't just one variable that distinguish these different cities so in an RTC we can actually claim look it's always this one thing in the elementary school example since we had a large data set and since it wasn't the kids themselves that made any kind of decision we can be reasonably certain that this is the key factor the time of the day that took a test but in this example what we can pay attention to is that was clearly cities themselves that could self select whether their own school should be sort of into one treatment or not and that also makes it clear that there could be lots of other things that we don't know about again we could go to the original study and find the true answer here but already in principle we could be a bit more uncertain about this so in any kind of so naturally occurring experiment whether it's an elementary school the Spanish flu trying to pay attention to who are these so what these people or schools or cities or companies that are in different treatment groups how did they get there and then a natural experiment it simply then defined by it was something that happened in nature and the elementary school example is then not quite clear on it wasn't research as it did it but but yes so an existence would showcase something that clearly is something we would call a natural experiment it's not quite a true randomized control trial in the sense that we can't be hundred percent certain that there wasn't some kind of systematic factor not a hundred percent certain like when we flip coin in a lab but we do notice here in this example that we saw and study good thing one that there was variation in the amount of nuclear fallout caused by the Chernobyl disaster and there was this regional

variation and the regional variation depended on rainfall it turns out that depending on how much rain a given area got it got more nuclear fallout and this leads to variation in standard I'd cognitive test 25 years later I have a longitudinal data set seems to be a treatment and the treatment is how much nuclear fallout do you get as a person living in Germany at that time as I said I lived in Germany there and 86 so it's not a formal randomization process but I think it's a situation where we can be fairly certain that this probably is as good as random the link between cognitive ability and variation in that and the and whatever happened sort of early on time we does seem like that we can say the Chernobyl disaster was a key factor the region's various areas in Germany they didn't decide himself whether they should get rain falling on it was an external shock it was an external factor which is why we can call it a natural experiment we would still usually call it some kind of quasi experiment because again we can't be absolutely 100% certain that there isn't some kind of relation between whether you're in agricultural area or whether or whether there could be any other systematic reason for why cognitive ability is linked to these different regions and the way it is so again natural experiment it's something that sort of happened outside the researchers control of in nature in this situation with the rainfall we can consider it basically as good as a random but we still do prefer in the proper randomized control trial to have absolute control of the situation and be certain this group or this treatment that group didn't get the treatment and there's only one difference between these groups

<https://youtu.be/L15ZJ8OmpQA>

a brief note on research designs that also includes some answers to questions about Bernstein cases and randomized control trials so we have these different types of research designs that we can identify and we know there are strengths and weaknesses associated with each of them cross sectional we could consider is like a picture it's it's a snapshot you have a picture of something right now this picture could entail asking kids about how many hours they played video games over the last three weeks so but it's still a snapshot of what they think right now it's not actually collecting multiple data points so just like a picture of cross-sectional and we have longitudinal with like a video camera or having multiple pictures taken you can see how stuff changes over time for instance then so you have multiple data points you've asked one months ago about you asked kids about how many hours week they played video games and now you're asking again today and you can now get the difference you're not just getting the perceived difference of what they think what they did 1 months 1 year 10 years ago you're getting insight into how their answers actually develop over time the experimental approach is then the the treatment someone gets the treatment the match that is on fire and the other one is not on fire and that leads to different kinds of outcomes and you would then of course have many different ice cubes you would do this on in order to see is this not just a random fluke in some sense you could say that it experiment is always about overtime if you start out with two identical groups and then you do something just something different to these two groups and see how that sort of what happens then after that but you can even as in the Bernstein situation have an experiment that runs over five

months and then see how the difference is develop over time
case study is then not just a picture
it's it's it's actually understanding something about a big specific case a specific context this could be one individual, accompany this could be a group of individuals group of people playing video games together for instance so a case is is so slightly different and more tricky to define but it just it focuses on the context it doesn't focus on the the variance within how thousand different people answer a survey it would focus on the particular people and their particular answers and why they gave these answers I should add that a case study can be an experiment as such we can have an experiment embedded in a case as in Bernstein there was a large Chinese factory and different units than either were behind a curtain or not but you can't run an experiment on one person on one sort of single firm then you would have to run it on people who lots of people within the firm has to be something you can compare to if you only have one individual you can't in terms of how for instance to address if there are multiple research designs other than actually two different studies we should assess differently and or separately and and I would say no I mean you an overall research study can consist of different research designs and you can then first identify well I think it's a field experiment first and foremost but has longitudinal elements that is actually also trying to understand a particular case but and then that's it the important point here is when you then assess the quality of the research design you shouldn't consider them in isolation yes there are some weaknesses related to the field experiments approach to answering whether something has an effect or not we don't really get insight into why it has an effect this is an the case based

interview approach could do and you have to assess the overall research study that first time carryout you have to combine these two things and you soli considered in isolation you can of course say the weaknesses of one thing one approach can then be alleviated can be sort of can be corrected so to speak by by the qualities of for instance here the case based interview approach so consider it sort of from an one overall one overall perspective of what's the strengths and the weakness of Bernstein's study identified that there are multiple research designs in play and then make that lead into an overall answer to the strengths and the weaknesses of the study that bench time carrier

<https://youtu.be/LQ6ZLZloG0o>

hi everyone I want to do a short video on causality and the challenge of identifying if there actually is a causal effect it's really been a key aim of this entire course that's why we've learned about experiments that's why we've learned about research designs in general and try to improve our ability to become better at identifying what can a study actually do because it's the aim of much of Sciences to be able to say look this kind of management style actually improves performance implementing a curtain in the Chinese mobile factory actually changes something so this has really been a key aim that we want to get how can I study create a causal or claim a causal effect and what does it mean to say that something is true or something is a true story about something and and really a key challenge is also that in the media and journal we tend to see studies being overblown exaggerated and this is actually what

what this slide is particularly about and showing a lot of stuff going on is let me try to explain it so we have the 50 studies that has been shared the most of all the studies published at this outlet they have generated 1.3 million social media shares so these are scientific studies that have been shared a lot on Facebook Instagram Twitter etc and what is then showing is that these the authors of this study they looked at the 50 studies and try to see how strong the cause of claim one could make so if you're all the way out here to the left well then it might be just any kind of correlation it might be small sample size there's really a substantial risk that there was not an opportunity to say that something causes something else and we can't say if the study claims that video games causes aggression or if it's just a small sample correlation and poor collection of data well it's just not a strong claim the further we go to the right the more likely that's what the authors of this article claimed more likely it is the causal inference is generally well conducted and reported might be some issues and that we can't be 100 percent sure but even over here in the very very high section that says causal inference is extremely likely to approximate the results of the ideal randomized control effect so so so even out here it's saying it's not necessarily perfectly true and the causal effect but at least we can approximate this is as good as it gets and as you can see most of these studies that have been shared like 1.3 million times actually only very few of them out here on the right three are in these two categories most are far further on the left actually don't allow the strong claim of look something causeas something else okay this is the categorization of these 50 articles then we look at what team articles of these

the authors of these 50 articles themselves claimed and it turns out a little bit more optimistic about their own results and who knows who who is right and then that's not the main there are more optimistic they think thinking so they have stronger claim to causality but the key thing is really to notice a difference between what the media says that's the bottom line the media says that 44% of all articles can claim I have a really strong claimed causality it says something like is caused by is due to statements without room for doubts or in general but even the moderate ones here is that well may be caused by seems to result in causality is suggested which is sent in the end 39 plus 24% almost all studies are quite strongly hinting at causality are claiming it even though if we go back and look at the original 50 articles it just doesn't seem to warrant such a strong claim the data does not seem to warrant the strong claim that we see here so in a nutshell that's actually key challenged that I want her to know about this is one of the key things that I wanted us to focus on and all about that this does happen this is a problem we see overblown causal statements in the media and I wanted us to become better at then understanding when something can be a causal effect and also which kind of research design that enables us better experiments field experiments longitudinal study carefully design correlation studies that might sort of at least have a better shot at not being absolutely on the left but can sort of push it to the right that's always what we want to do push it to the right on this overview because although and also indicate before we should not think of causality as a binary thing it's not just true or false it's some claims are stronger than others it's not true or

false as such whether having a darker skin color leads to a higher risk of getting a red card but this explanation is certainly getting us there the internal validity is fairly strong but what we should also think of the explanation the causal explanation in terms of external validity it might hold true for these European football leagues you don't know if it also holds true for all kinds of other sports or in the future etc but and then also related to this one might go away from from what we've heard so far and say ok so apparently we need a randomized control trial to really be able to say that something causes something else and and and true it's a golden standard it's something that really enables us to get there but we shouldn't think of it as the only way of getting there and let's think of this public private school example that we had in a study cafe I won't go through it in detail here just want to remind you this is the study that we're talking about does putting a kid in private school improve their performance and we saw this over here where the claim is does putting kids either in public private school improve the academic achievement and and and it was then clear to us when we assisted we don't know if the children's ability before they even start school this different so it's actually the children's ability that then decides while clever kids get into private school and and and clever kids also get the higher academic achievement so that's an alternative explanation that it turns out to be the cognitive ability that decides it doesn't really matter which school they go to and in addition we also talked about well maybe it's the parental resources it's it's focusing and having the resources to be able to study at home maybe it's the the values that are being

put on education maybe parents that sent the kids to private school just value education much more which then means to end up performing better it's not the school type that makes it perform better it's just because these two things are correlated so again the conclusion could sort of be okay apparently we need a randomized control trial do we need a thousand kids that we randomly put into a public private school and look at them ten years later that would be quite something but if we now collect data on cognitive ability the resources of parents how high education is valued let's say we actually collect some kind of survey data on that then we test the kids ability before they start school we test and question how much parents value education and we have this information and control for this information in the statistical analysis well in the end we could then show if and how school type has an effect on academic achievement we don't have a randomized control trial so so in principle there might be other things that we should have thought of but at least we can in a careful design study where we have data over time it can try to get closer to a conclusion this is really what we want to become better at being sure than a scientific claim is credible so if we've done all this kind of data collection we can be more sure of our conclusion not necessarily on a person sure but that's all we can do in science is to sort of push our certainty and reduce our uncertainty and that's what we do here this particular example